# Homemade Memristor

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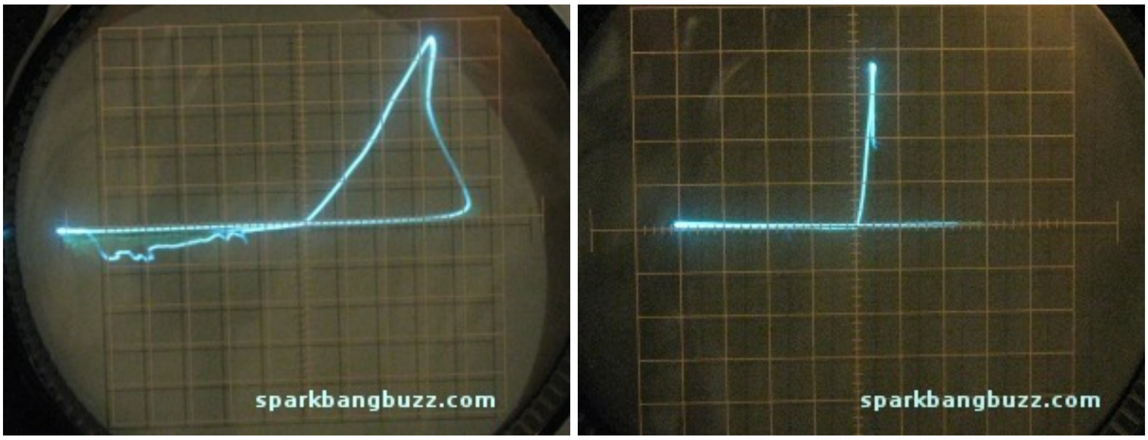
## Curve traces from two homemade memristors.




**Modified curve tracer is applying AC voltage. Horizontal axis represents voltage and the vertical axis represents current. In both cases, the curve always passes through zero voltage and current. This requirement must be met in order to be classified as a memristor.**

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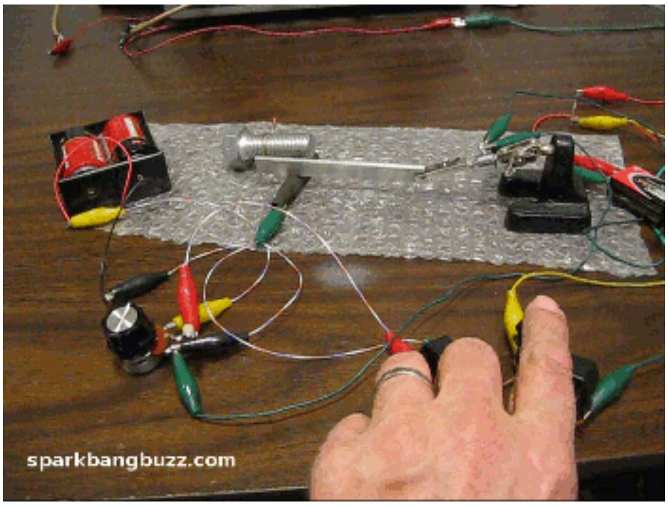
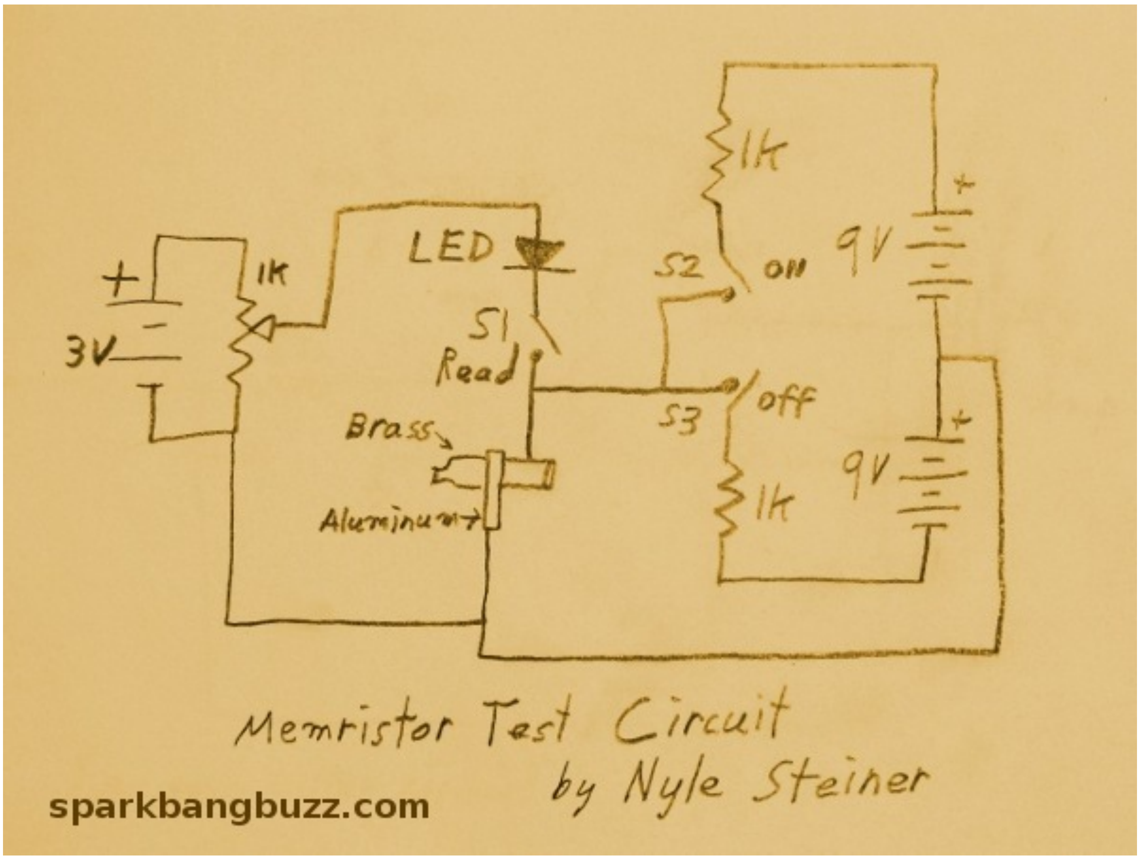
## Memristor test setup and schematic.




**When S1 is closed the LED shows the status of the memristor. If the memristor is in the low resistance (on) state the led will light. If the memristor is in the high resistance state (off) the led will not light. S2 turns the memristor on. S3 turns the memristor off.**

**The 3 volt battery and 1k pot supply just enough voltage to light the led without effecting the on off state of the memristor.**


**Nearly ten years ago I was hiking up a mountain road when I noticed a lot of small gravel and the smell of sulfur. Some empty brass cartridges that were lying in the gravel had turned to a dark black color.**


**I took some of the cartridges home to investigate the electrical properties of the dark black corrosion. When the cartridge was contacted with a piece of aluminum and connected to the curve tracer, the left pattern shown near the top of this page was**

**observed.**

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**I let several pieces of copper, brass and lead set in a container of sulfur for a considerable length of time.**


**All of the above pieces, after removing from the sulfur, displayed similar memristor**

**characteristics.**

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**I recorded this in my notebook as a curious memory phenomenon but did not think much more about it until recently reading about the memristor and realizing how similar it is to what I had been observing. I decided to do some additional experimenting and build the test circuit (animated photo and schematic shown above) to demonstrate the memristor action of the corroded pieces of metal.**


**I also tried putting a small pile of sulfur on a piece of sheet copper.**


**After just a few hours, I removed the sulfur and noticed that it had formed a black colored area on the copper. This black copper when in contact with a piece of aluminum produced the curve trace shown on the right near the top of this page.**

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